

Image Segmentation

Computer Vision: CS 566

Computer Science

University of Wisconsin-Madison

Image Segmentation

Group pixels with similar visual characteristics.

Topics:

- (1) Segmentation and Grouping in Humans
- (2) k-Means Segmentation
- (3) Mean Shift Segmentation
- (4) Graph Cut Segmentation

Active Contours

Given: Approximate boundary (contour) around the object

Task: Evolve (move) the contour to fit exact object boundary



Image

Active Contours:

Iteratively “deform” the initial contour so that:

- It is near pixels with high gradient (edges)
- It is smooth

What About Natural Scenes?

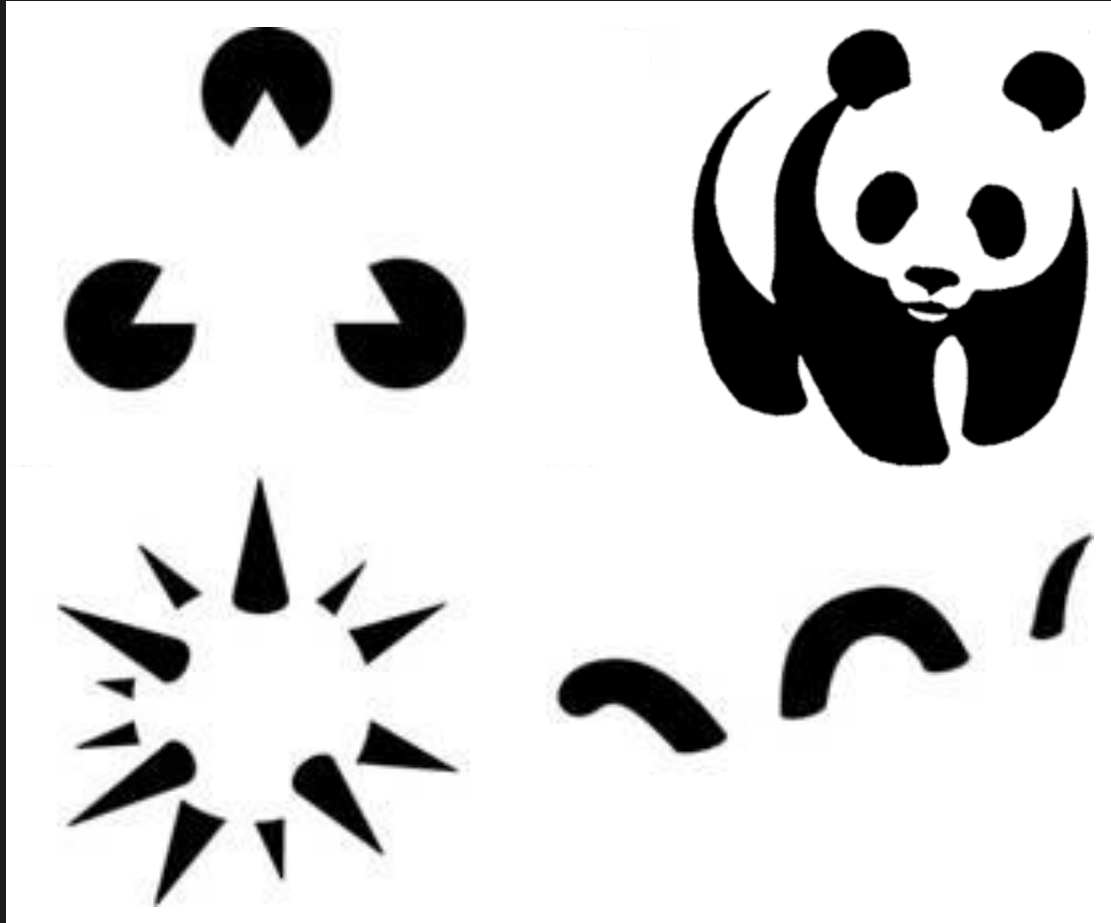


Past Experience and Emergence



Complex pattern formation from prior knowledge

Reification

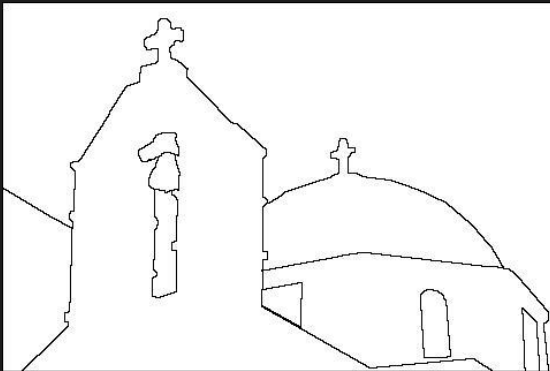


Illusory or subjective contours are perceived

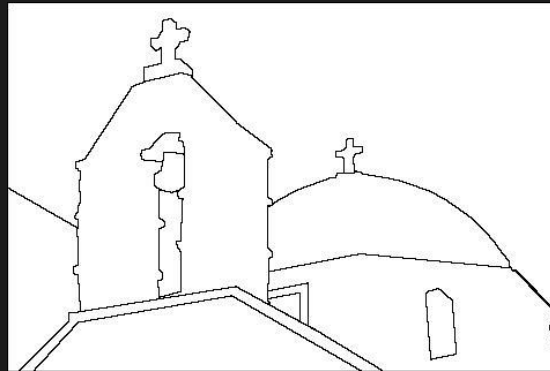
Image Segmentation by Humans



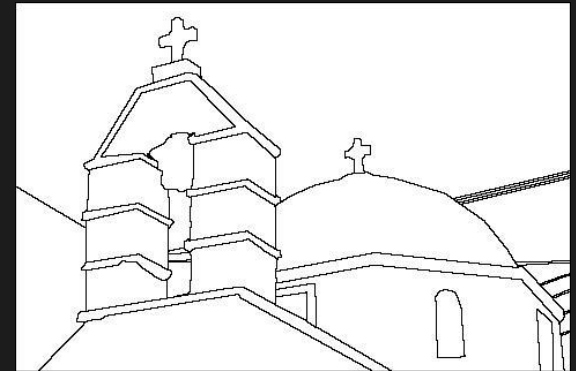
Input Image



User 1



User 2



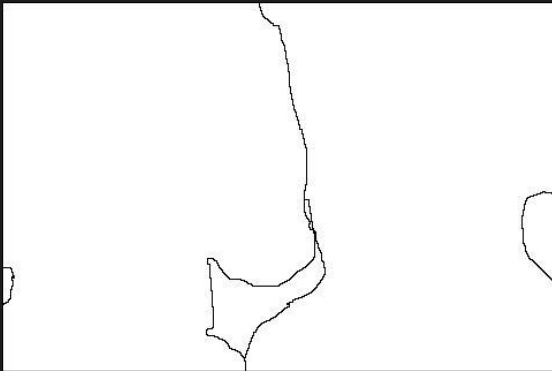
User 3

Segmentation is Subjective

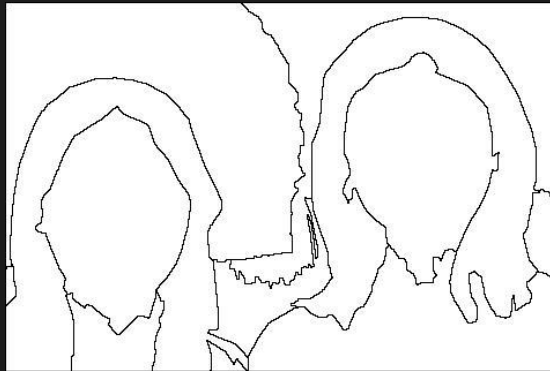
Image Segmentation by Humans



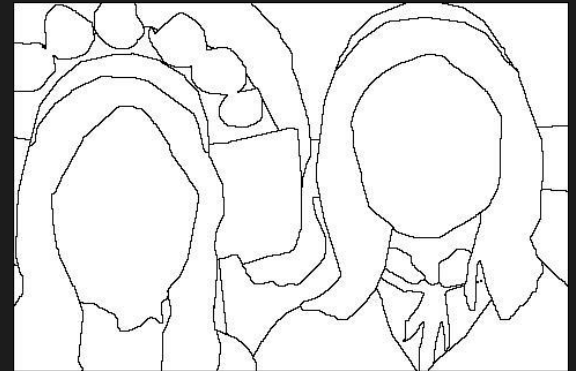
Input Image



User 1



User 2



User 3

Segmentation is Subjective

Segmentation Strategies

Segmentation is intuitive for us.

Very hard to translate these intuitions to an algorithm.

Top-down Segmentation: Pixels belong together because they come from same object.

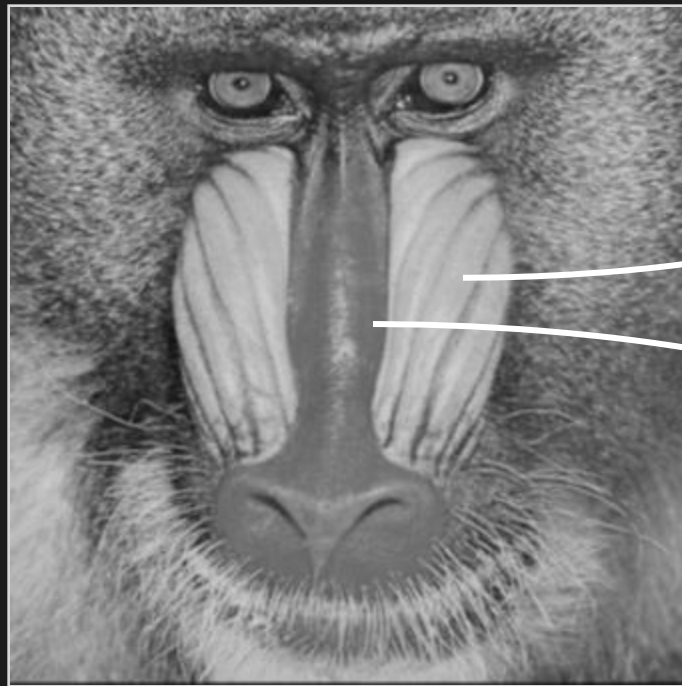
Bottom-up Segmentation: Pixels belong together because they look similar.

Visual Characteristics of Pixels

Visual similarity can be based on:

- Brightness
- Color
- Texture
- Position
- Depth
- Motion
- $\vdots$

Feature Vectors for Pixels



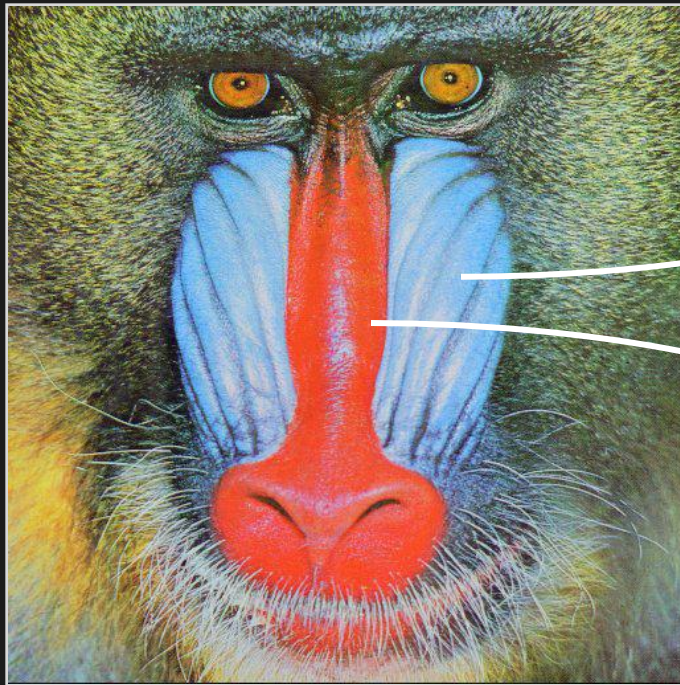
Input Image

feature vector
[154]

feature vector
[67]

Pixels as feature vector: $[I]$

Feature Vectors for Pixels



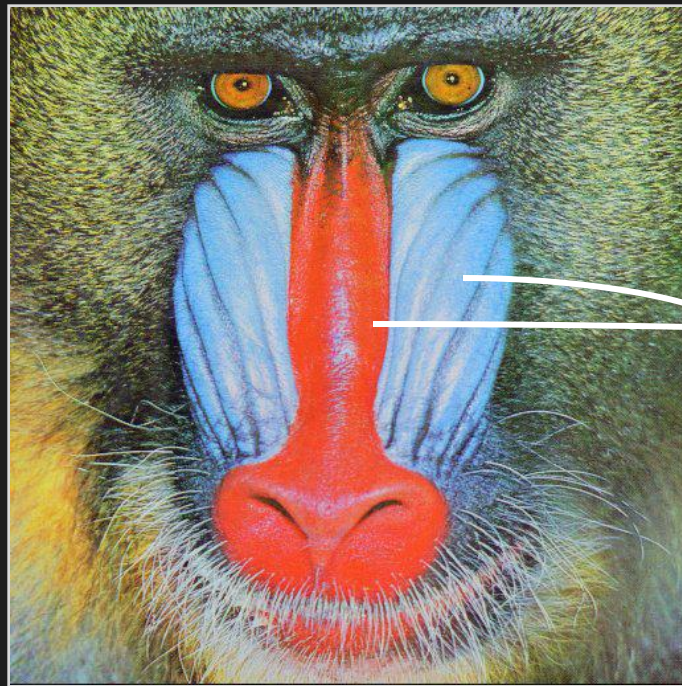
Input Image

feature vector
[62, 170, 230]

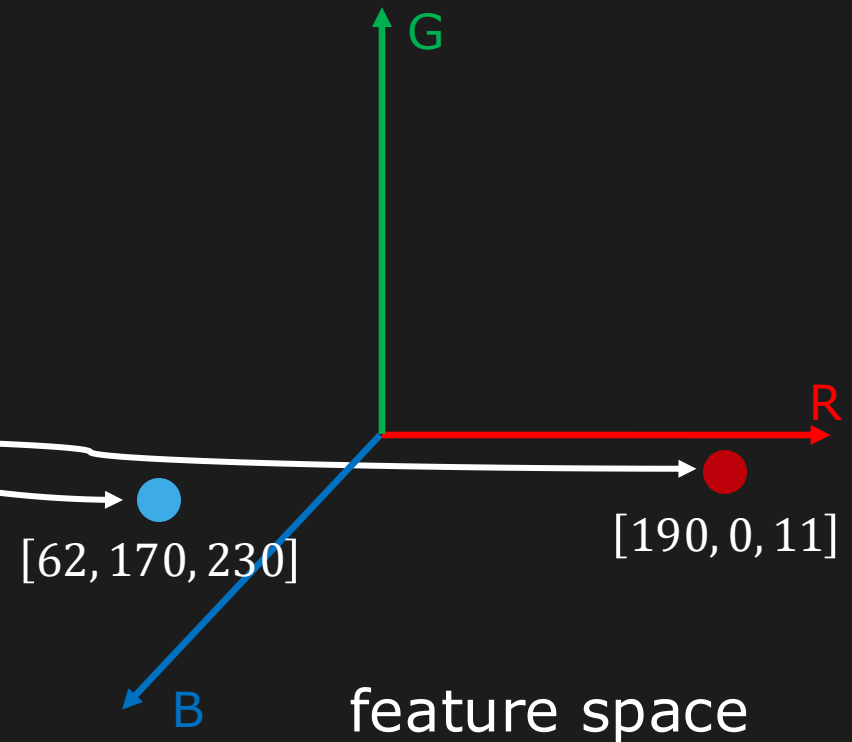
feature vector
[190, 0, 11]

Pixels as feature vector: $[R, G, B]$

Feature Vectors for Pixels

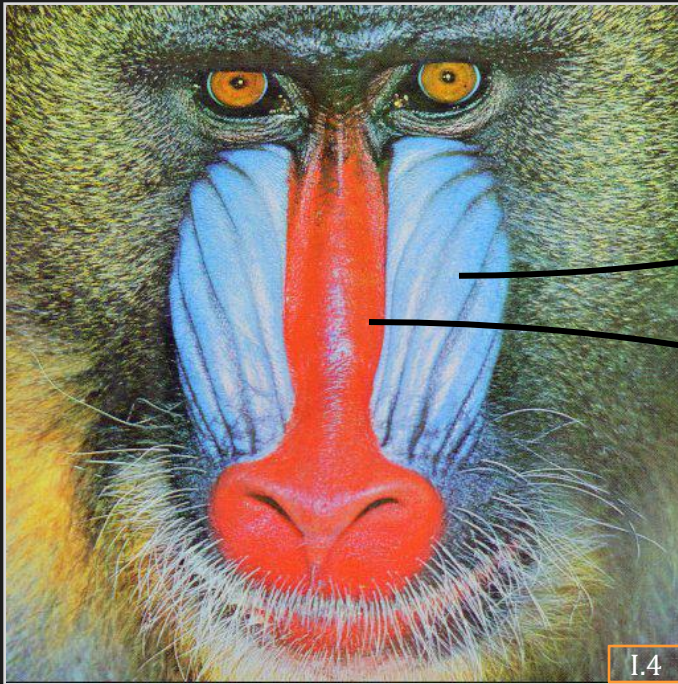


Input Image

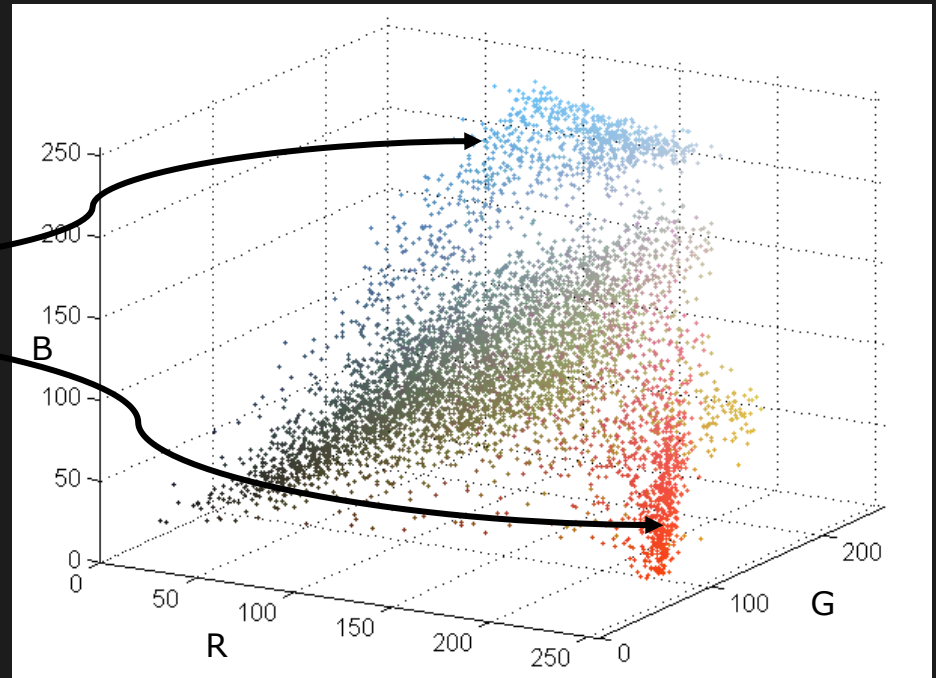


Pixels in Euclidean Space

Euclidean Space: Generalization of 3D Cartesian space to higher dimensions.



Input Image



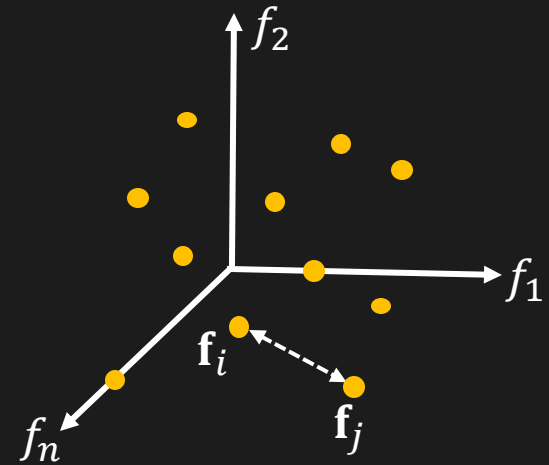
Pixel RGB Color Distribution
(Color of feature point $\equiv$ Color of image pixel)

Pixel Similarity

Let i and j be two pixels
whose features are $\mathbf{f}_i$ and $\mathbf{f}_j$.

$\mathcal{L}^2$ Distance between $\mathbf{f}_i$ and $\mathbf{f}_j$:

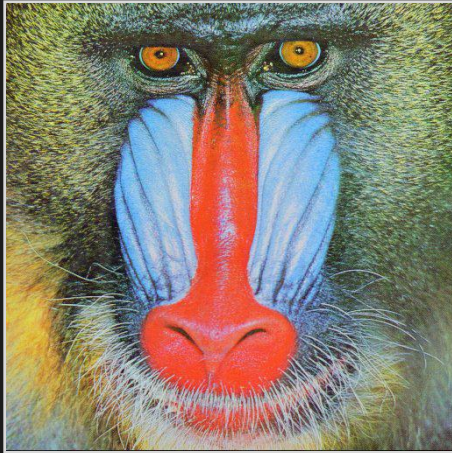
$$S(\mathbf{f}_i, \mathbf{f}_j) = \sqrt{\sum_k (f_{ik} - f_{jk})^2}$$



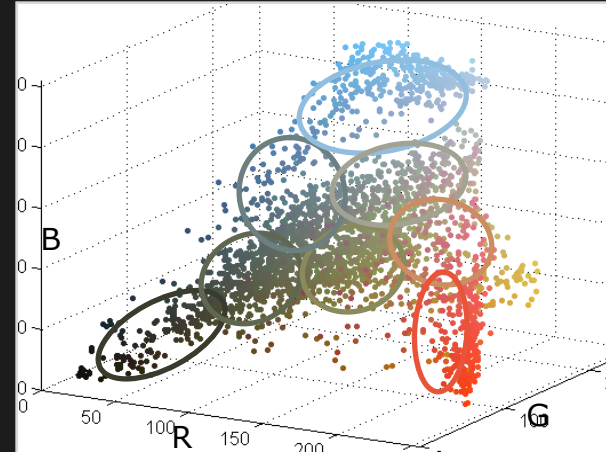
$[R, G, B, x, y, \dots]$ - space

Smaller the Distance, Greater the Similarity

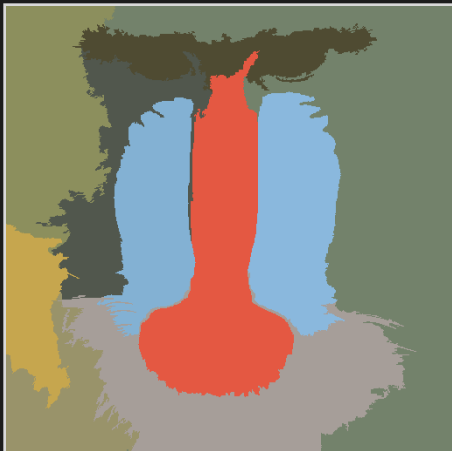
Clustering Similar Pixels



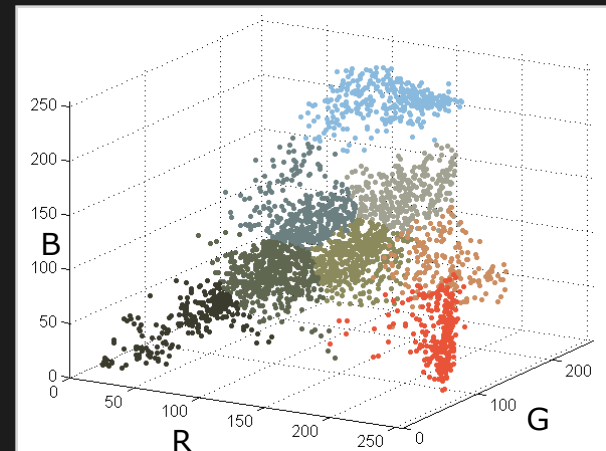
Input Image



Pixel RGB Color Distribution
(Color of feature point $\equiv$ Color of image pixel)



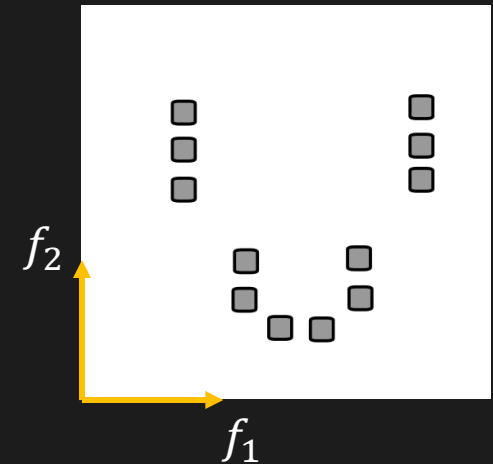
Segmented Image



Color-Coded Clusters
(Color of feature point $\equiv$ Color of image segment)

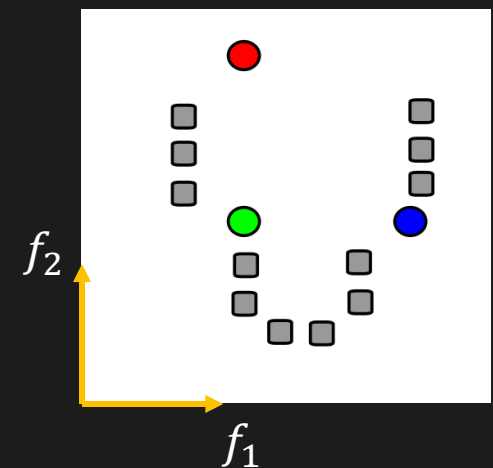
3-Means Clustering Example

Problem: Segment the given pixel feature distribution into 3 clusters.



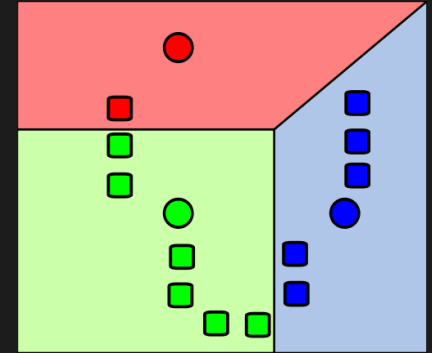
Solution:

Step 1: Randomly generate the initial centroids (means) of the 3 clusters.

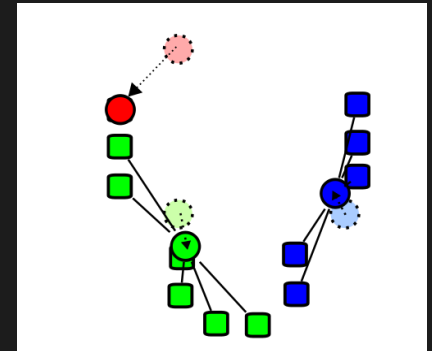


3-Means Clustering Example

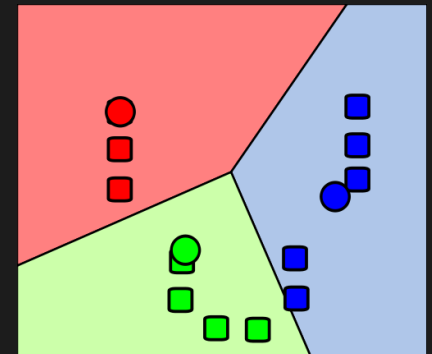
Step 2: Create 3 clusters by assigning each feature point to the nearest mean.



Step 3: Recompute the mean of each cluster.



Step 4: Repeat steps 2 and 3 until convergence.



k-Means Clustering

Given: Image with N pixels and number of clusters k .

Task: Find the k clusters.

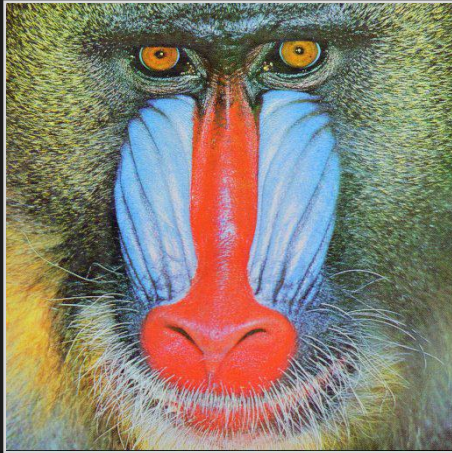
Clustering:

- 1:** Pick k points randomly as the initial centroids (means) $\{\mathbf{m}_1, \mathbf{m}_2, \dots, \mathbf{m}_k\}$ of the k clusters in feature space.
- 2:** For each pixel $\mathbf{x}_j$ find nearest cluster mean $\mathbf{m}_i$ to pixel's feature $\mathbf{f}_j$ and assign pixel to cluster i .
- 3:** Recompute mean for each cluster using its assigned pixels.
- 4:** If changes in all k means is less than a threshold ε , stop. Else go to step 2.

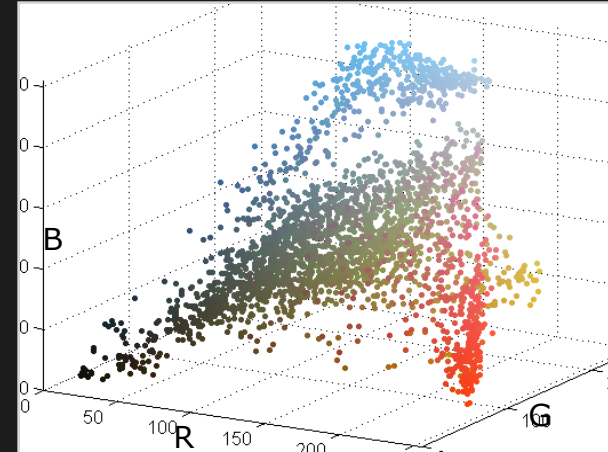
k-Means Initialization Methods

- **Method 1:** Select k random feature points as initial centroids. If two points are very close, resample.
- **Method 2:** Select k uniformly distributed means within the range of the distribution.
- **Method 3:** Perform k-means clustering on a subset of pixels and use the result as the initial means.

k-Means Clustering Results ($k = 2$)



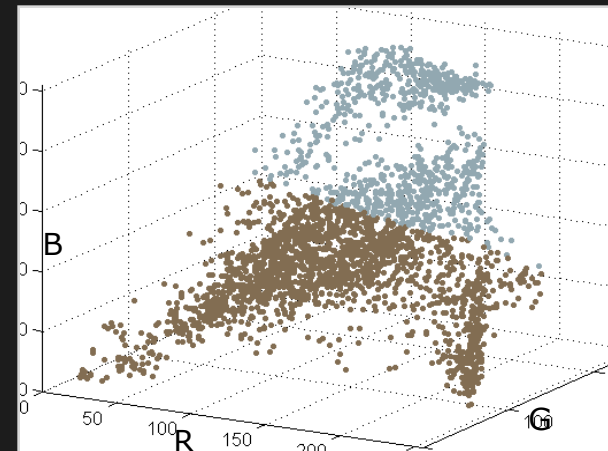
Input Image



Pixel RGB Color Distribution
(Color of feature point $\equiv$ Color of image pixel)

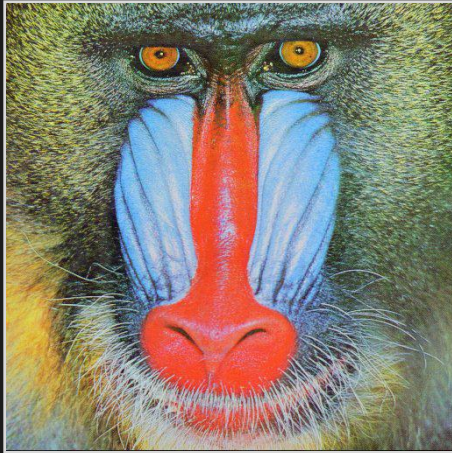


Segmented Image
($k = 2$; $\{R, G, B\}$ -space)

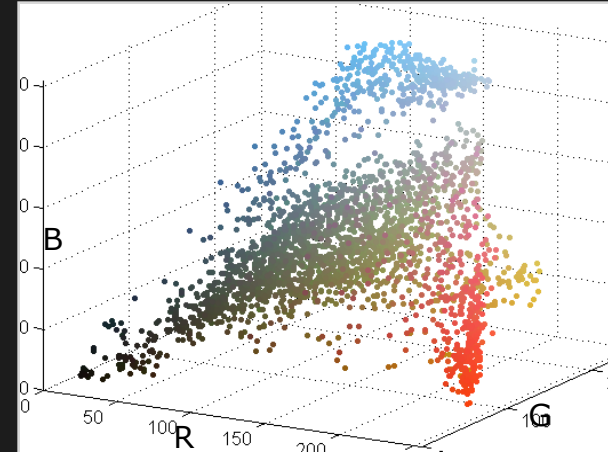


Color-coded Clusters
(Color of feature point $\equiv$ Color of image segment)

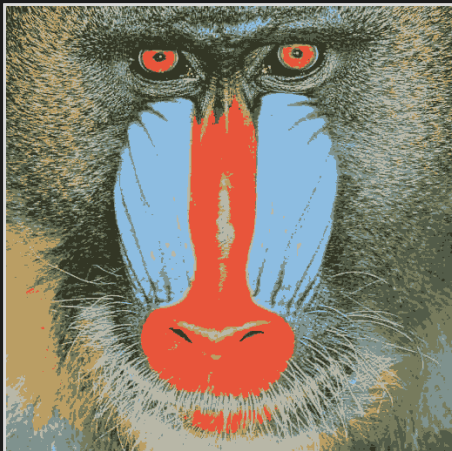
k-Means Clustering Results ($k = 8$)



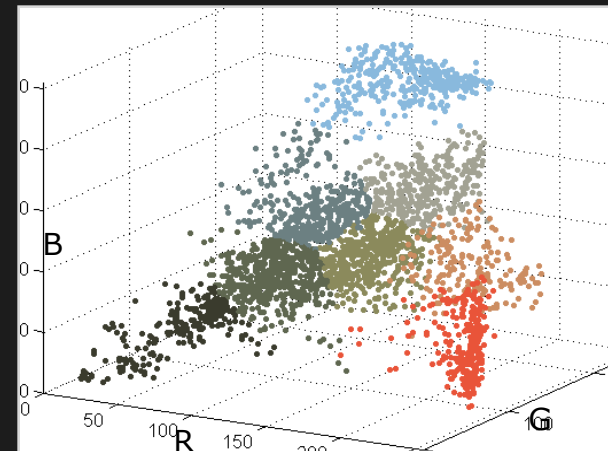
Input Image



Pixel RGB Color Distribution
(Color of feature point $\equiv$ Color of image pixel)



Segmented Image
($k = 8$; $\{R, G, B\}$ -space)

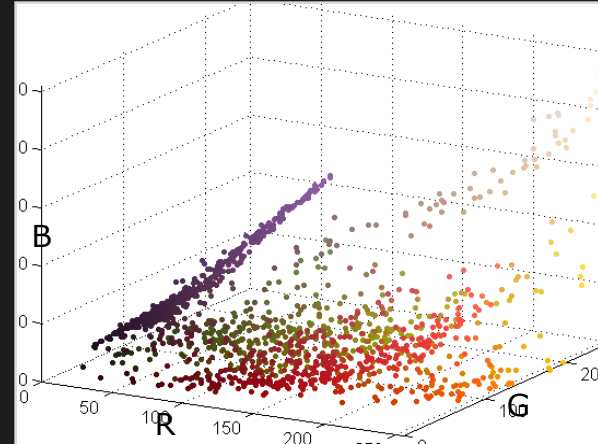


Color-coded Clusters
(Color of feature point $\equiv$ Color of image segment)

k-Means Clustering Results ($k = 16$)



Input Image



Pixel RGB Color Distribution



Segmented Image
($k = 16$; $\{R, G, B\}$ -space)



Segmented Image
($k = 16$; $\{R, G, B, x, y\}$ -space)

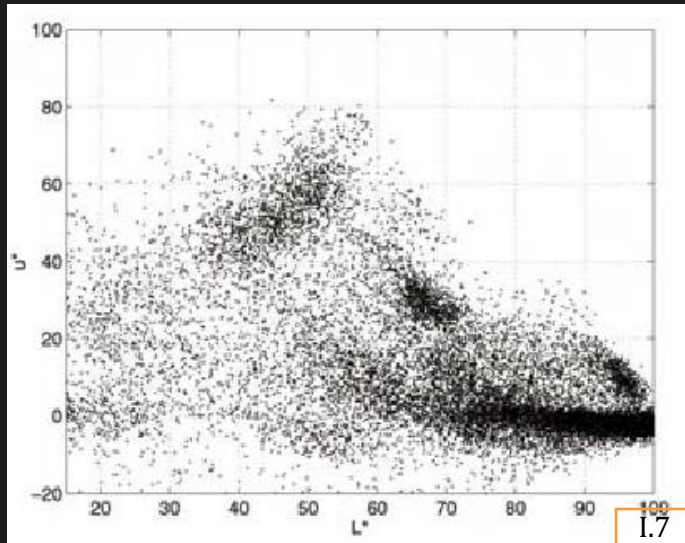
Note: Disjoint regions could belong to a single cluster.

k-Means Clustering: Comments

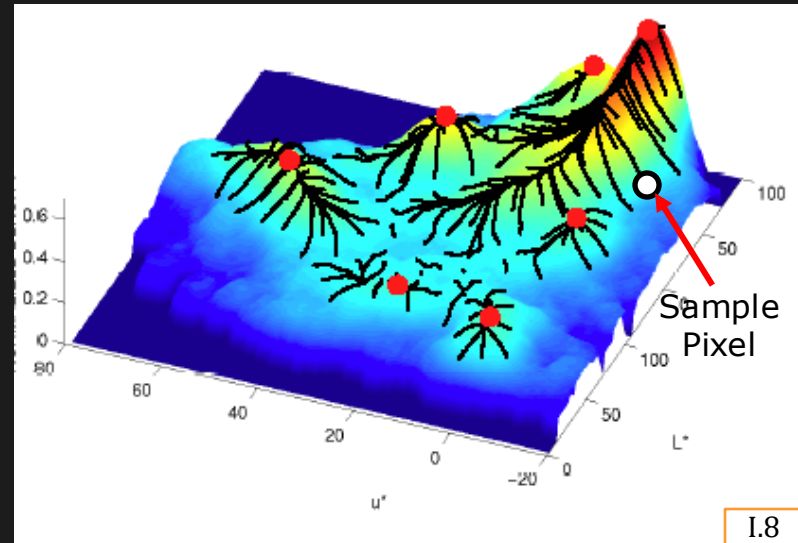
- Simple and reasonably fast
- Need to pick the number of clusters k
- Sensitive to initialization
- Sensitive to outliers

Can we do better?

The Concept of Mean Shift



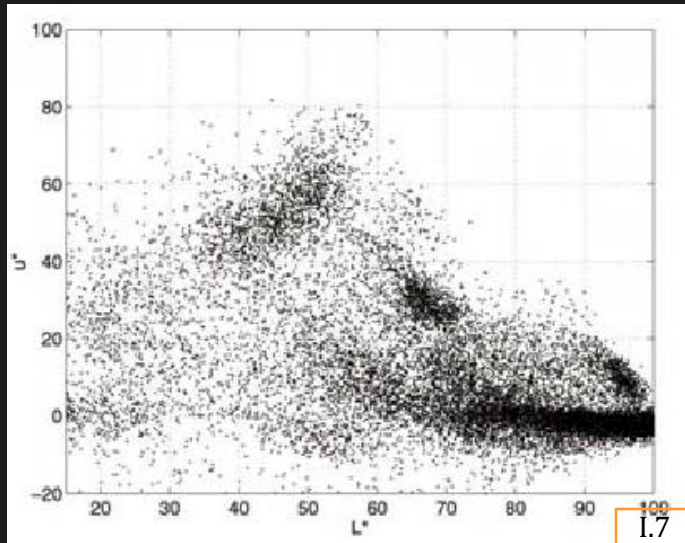
Pixel Feature Distribution



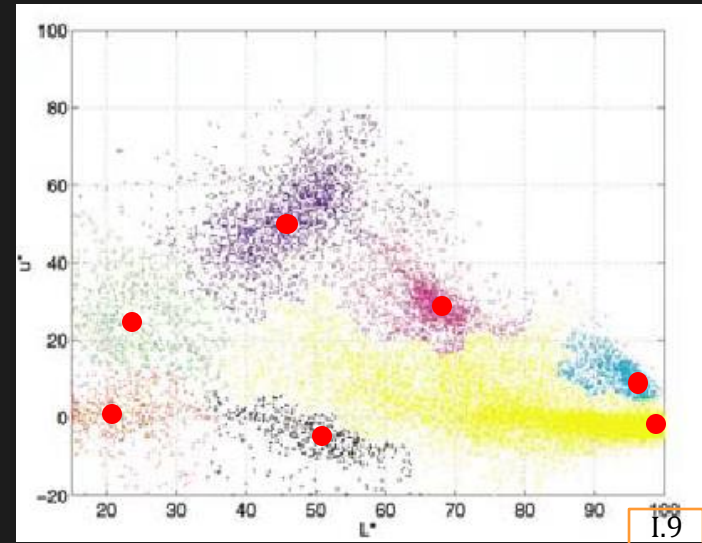
Normalized Density

- Each hill represents a cluster.
- Peak (mode) of hill represents center of cluster.
- Each pixel climbs the steepest hill within its neighborhood.
- Pixel assigned to the hill (cluster) it climbs.

The Concept of Mean Shift



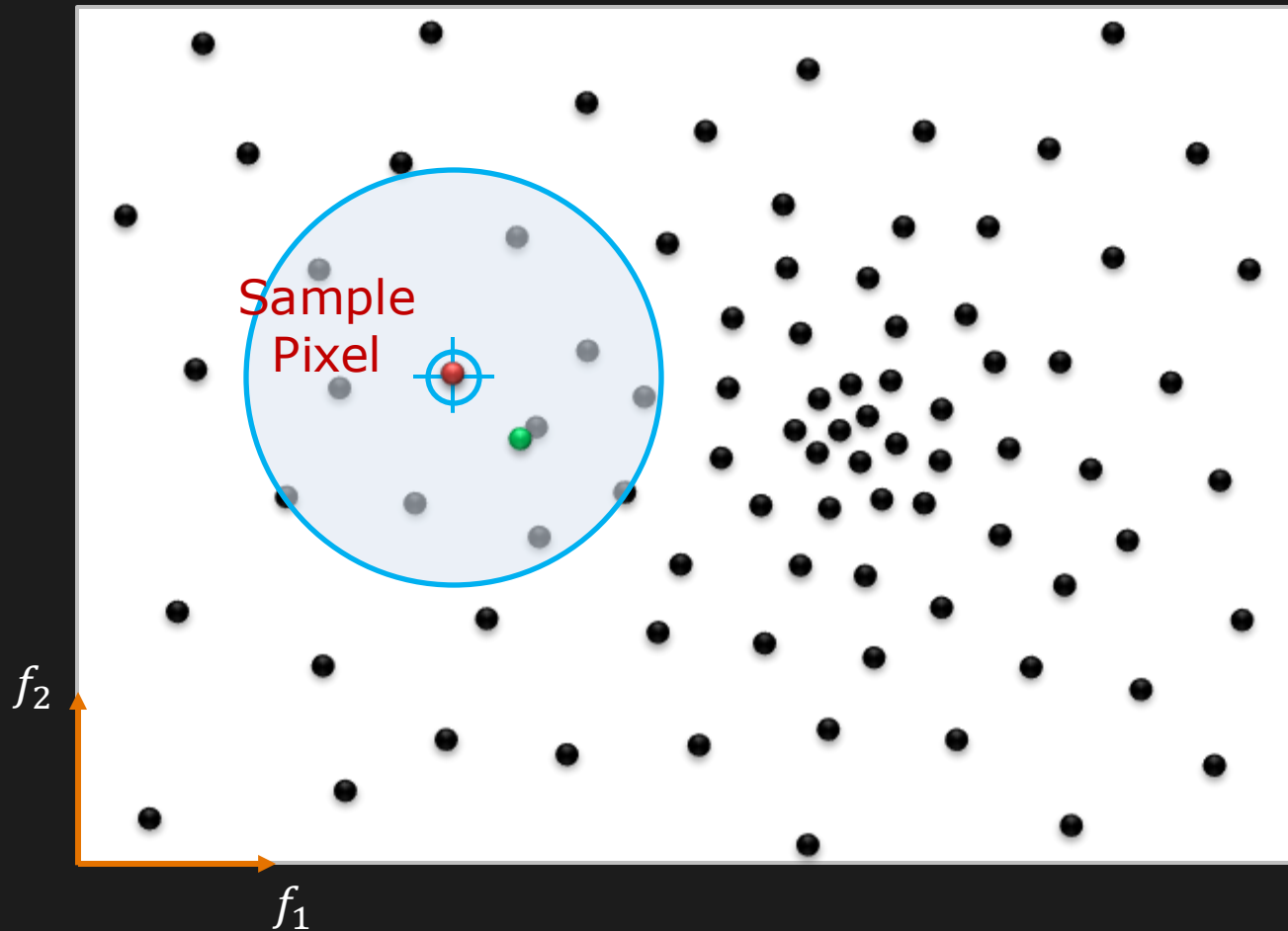
Pixel Feature Distribution



Labeled Clusters and their Centers

- Each hill represents a cluster.
- Peak (mode) of hill represents center of cluster.
- Each pixel climbs the steepest hill within its neighborhood.
- Pixel assigned to the hill (cluster) it climbs.

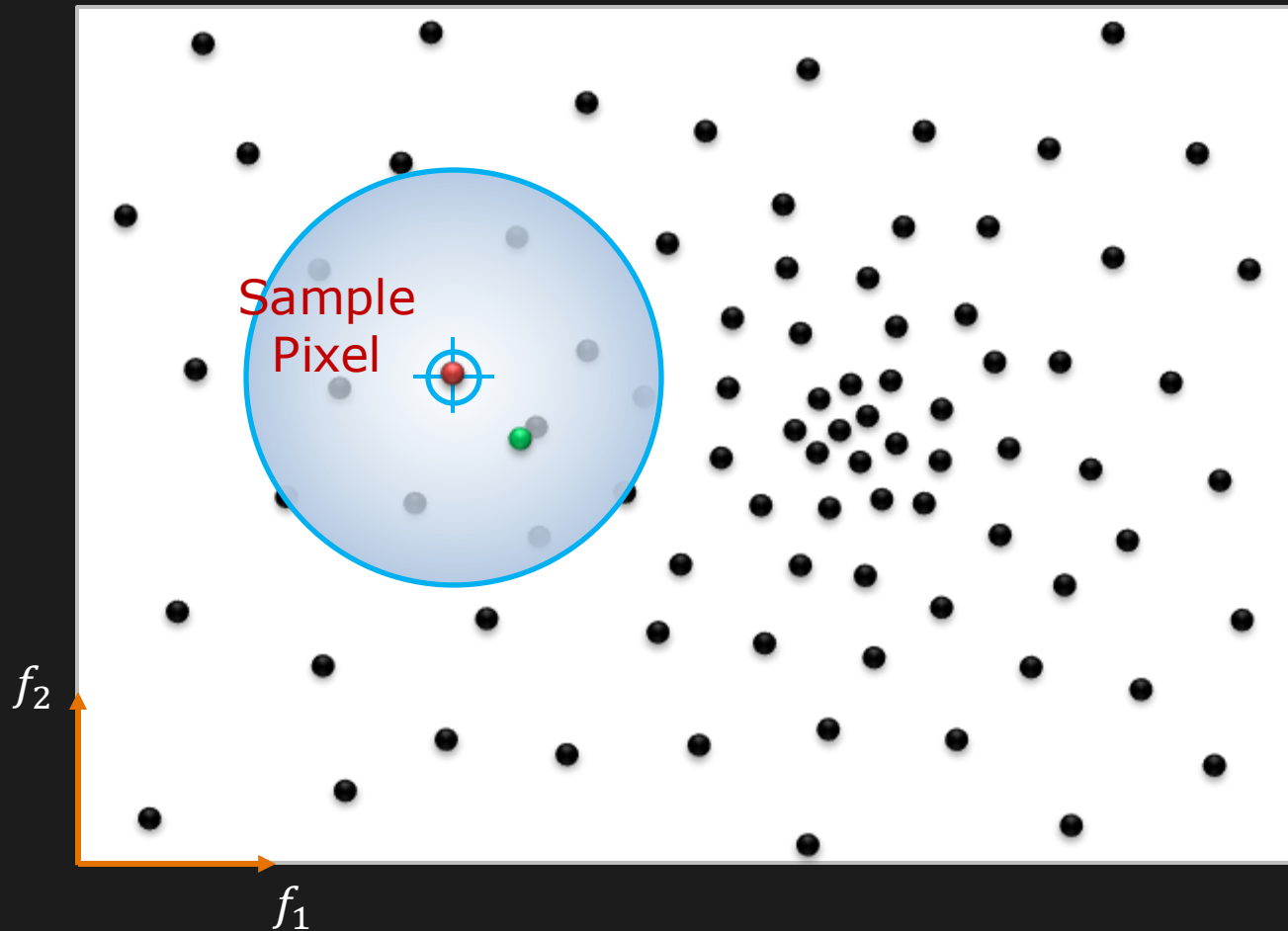
Hill Climbing using Mean Shift



Compute Centroid (●) within a Window of size W .

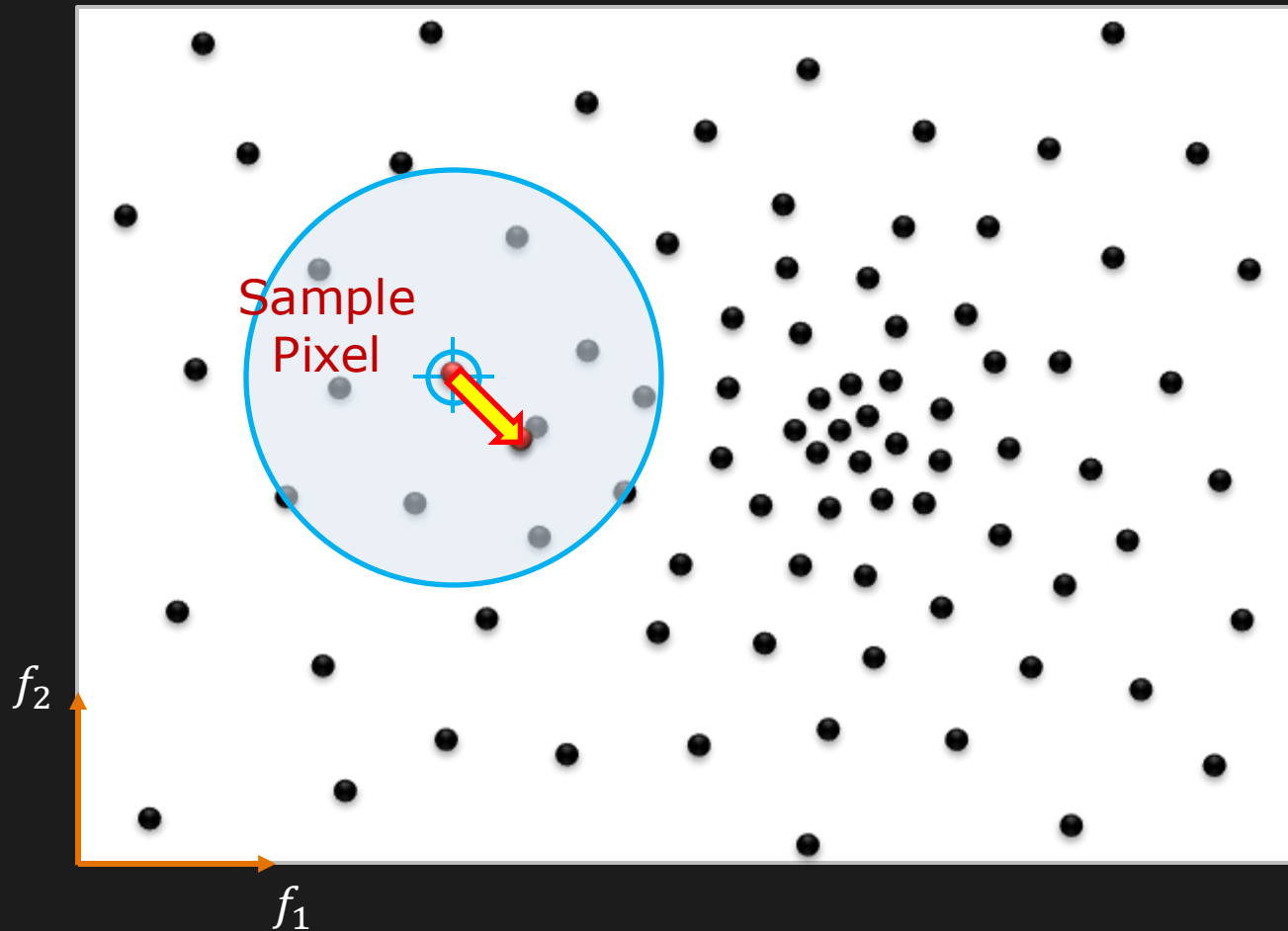
Use **Simple Means** or...

Hill Climbing using Mean Shift



Compute Centroid (●) within a Window of size W .
Use **Weighted Means**.

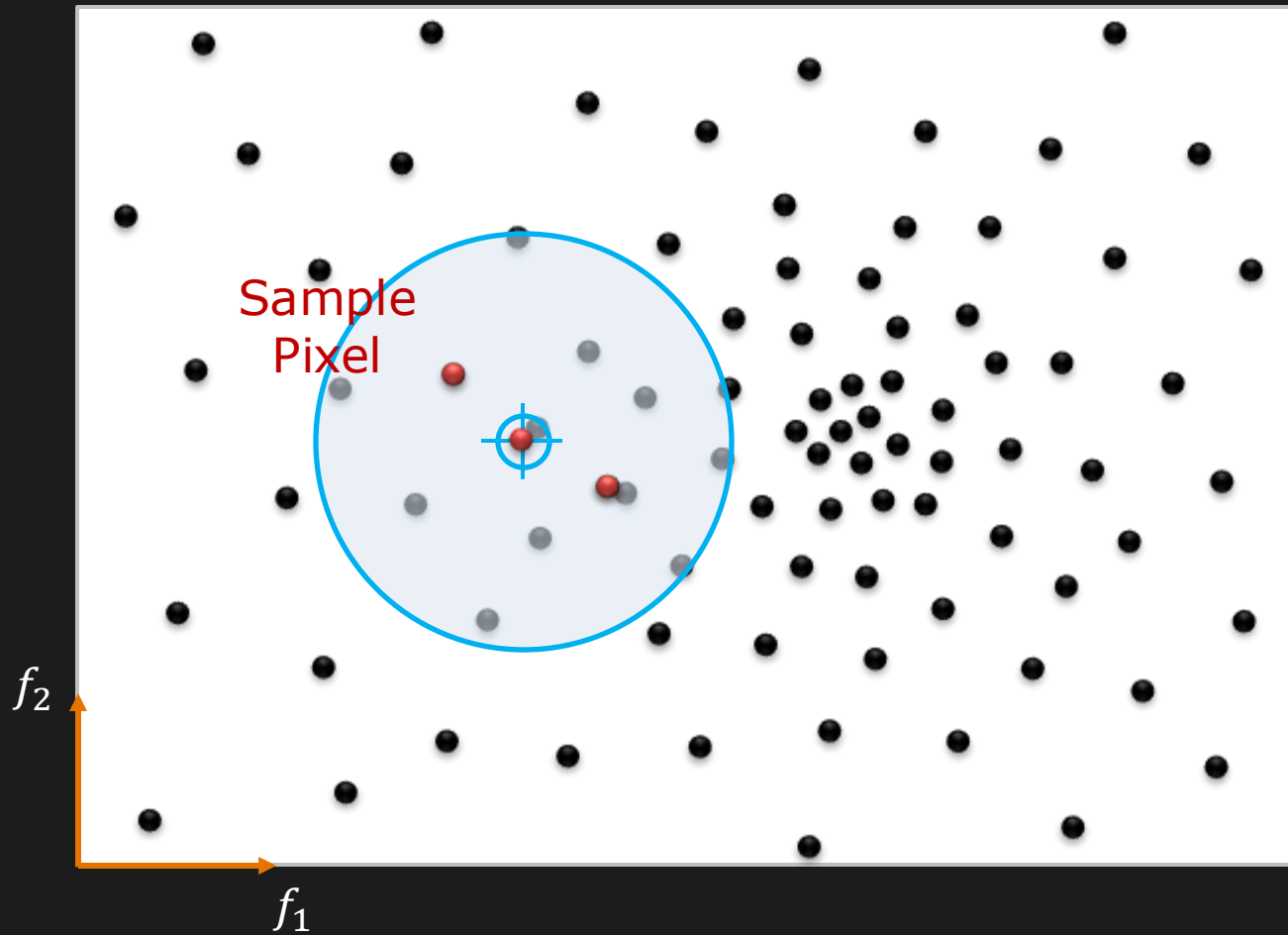
Hill Climbing using Mean Shift



“Shift” the Window to the Centroid

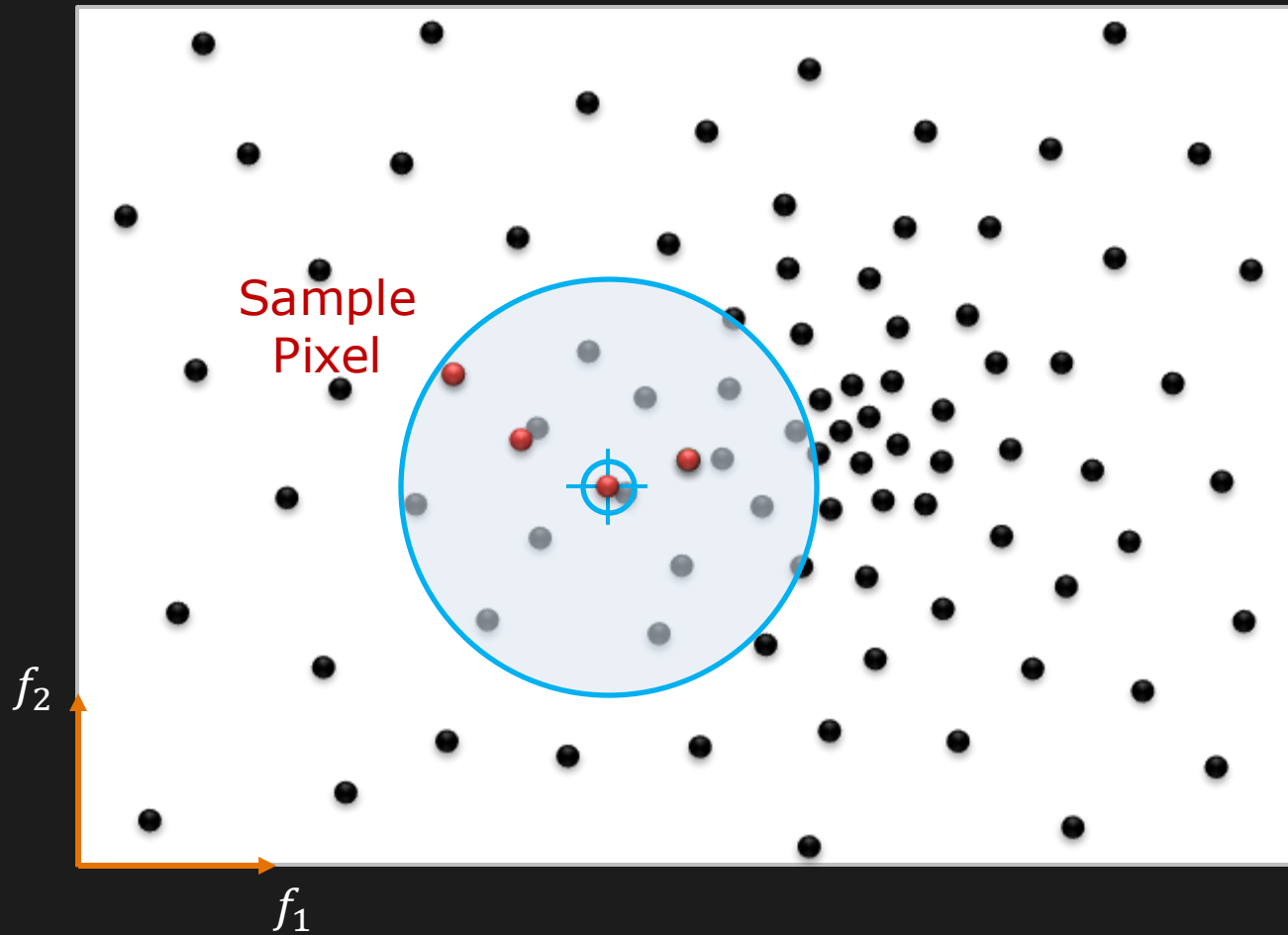
 : Mean Shift Vector

Hill Climbing using Mean Shift



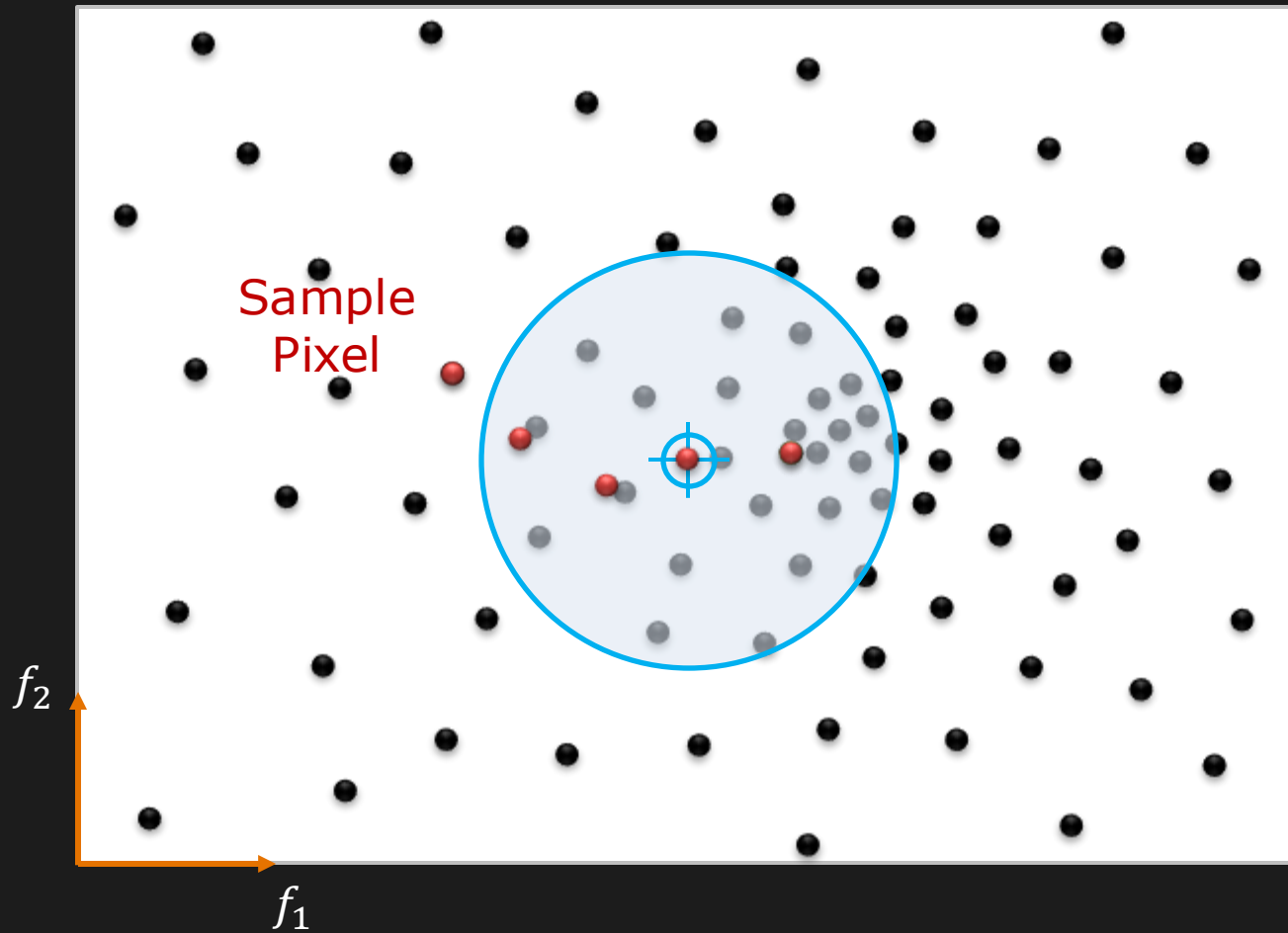
Repeat...Until Convergence.

Hill Climbing using Mean Shift



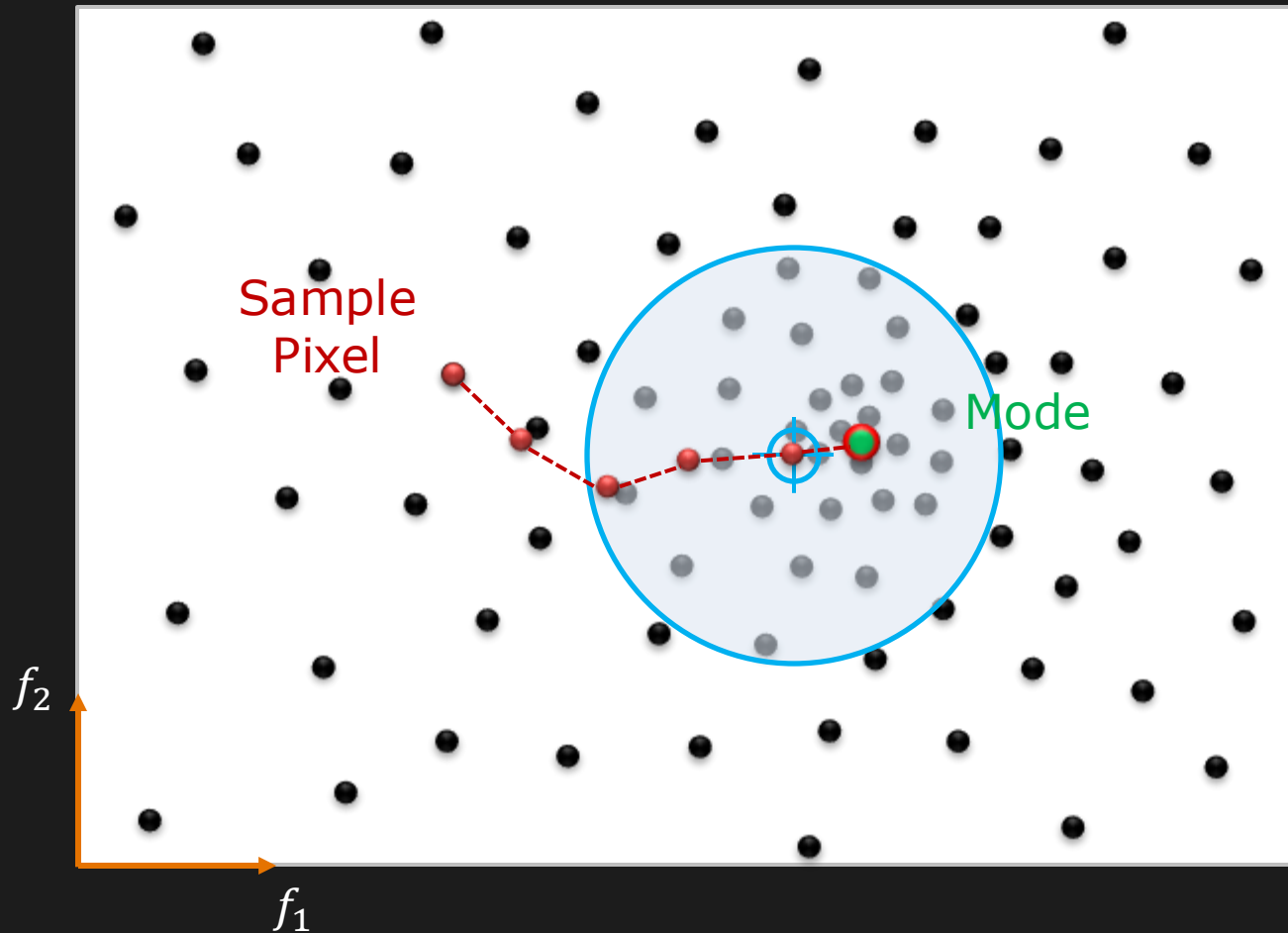
Repeat...Until Convergence.

Hill Climbing using Mean Shift



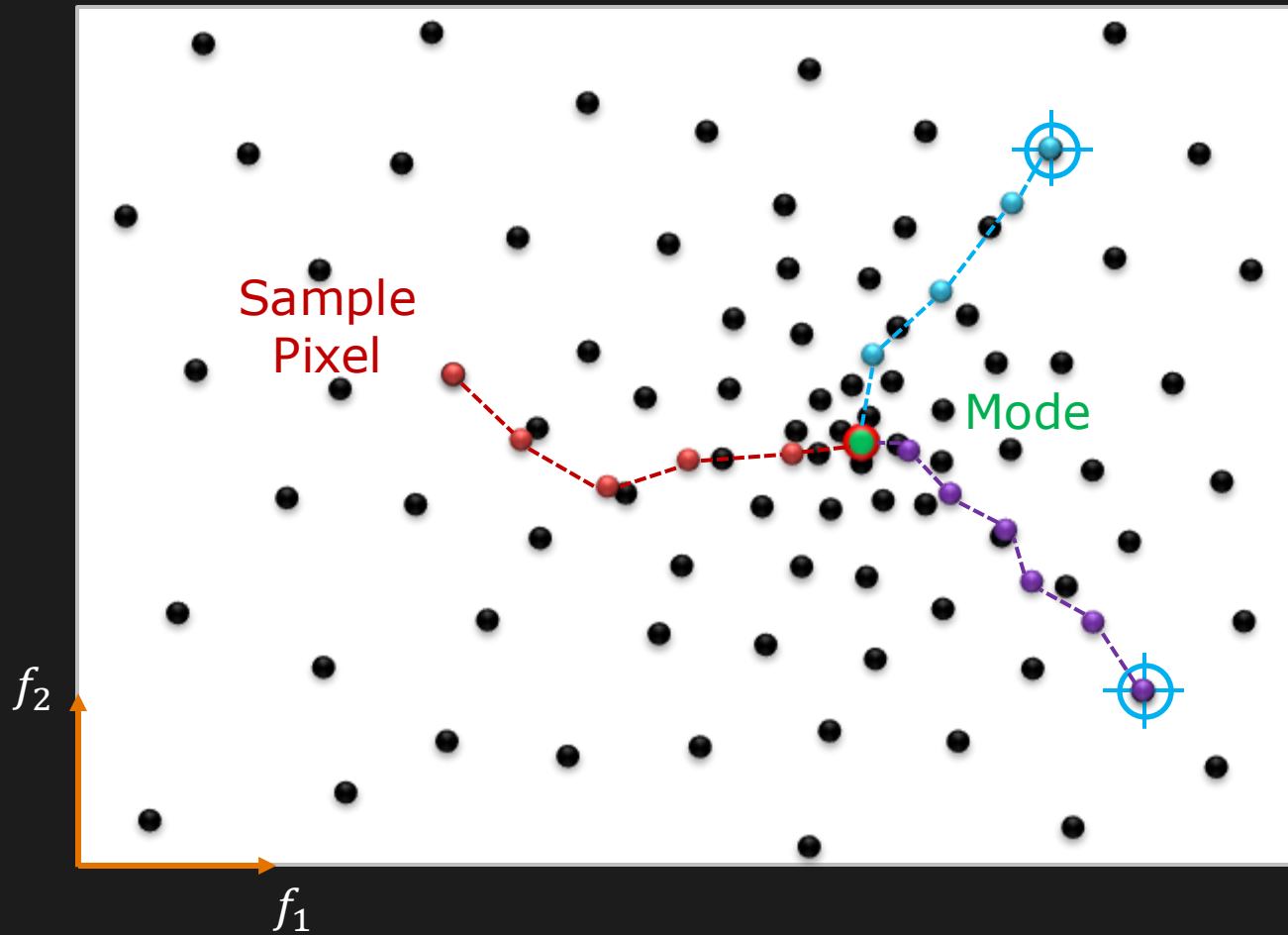
Repeat...Until Convergence.

Hill Climbing using Mean Shift



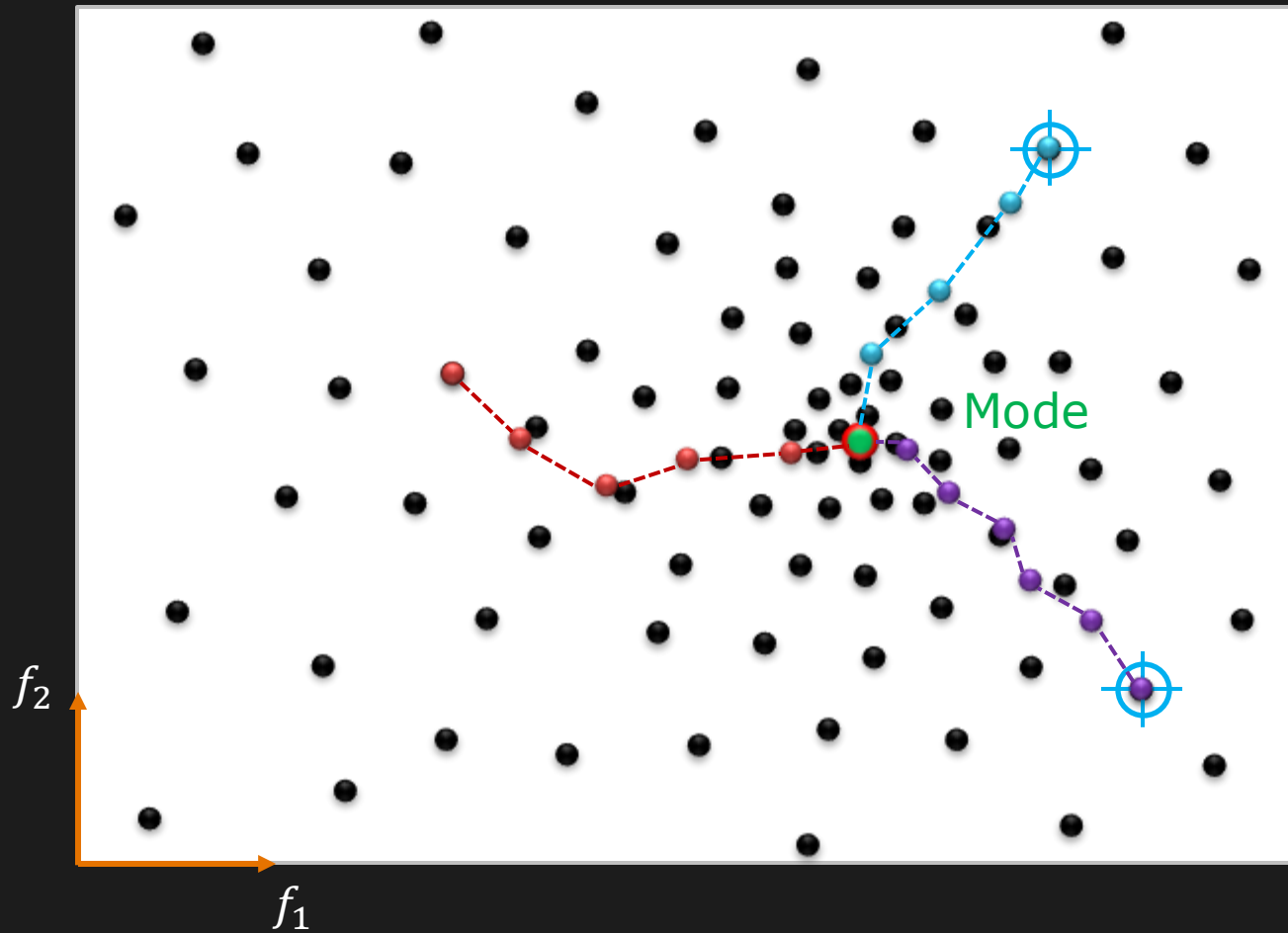
Declare mode and assign it a cluster label.

Hill Climbing using Mean Shift



Repeat...for all pixels!

Hill Climbing using Mean Shift



Features that converge to same mode belong to same cluster.

Means Shift Algorithm

Given: Distribution of N pixels in feature space.

Task: Find modes (clusters) of distribution.

Clustering:

1: Set $\mathbf{m}_i = \mathbf{f}_i$ as initial mean for each pixel i .

2: Repeat the following for each mean $\mathbf{m}_i$:

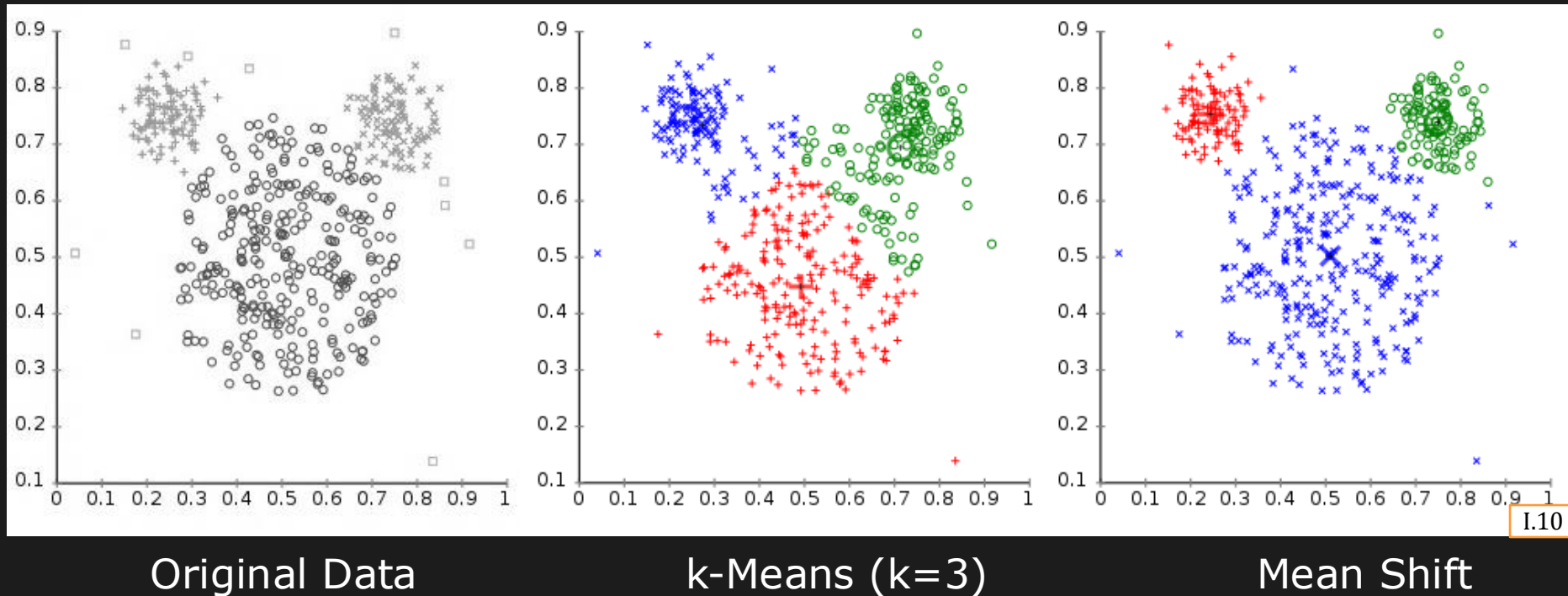
a: Place window of size W around $\mathbf{m}_i$.

b: Compute centroid $\mathbf{m}$ within the window. Set $\mathbf{m}_i = \mathbf{m}$.

c: Stop if shift in mean $\mathbf{m}_i$ is less than a threshold ε .
 $\mathbf{m}_i$ is the mode.

3: Label all pixels that have same mode as belonging to same cluster.

k-Means vs. Mean Shift



k-Means vs. Mean Shift



Input Image

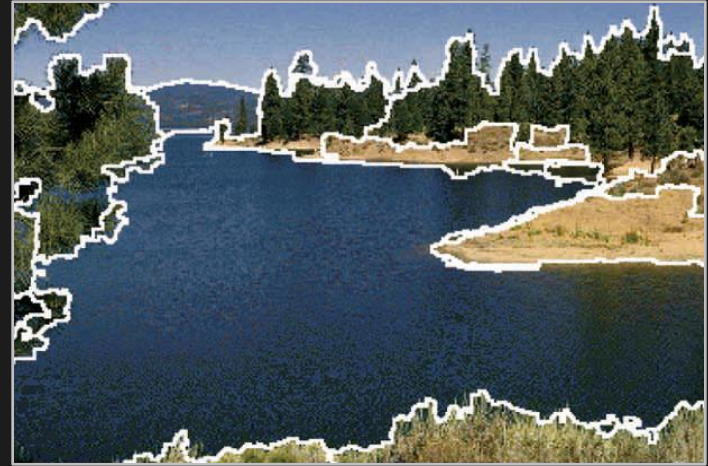


k-Means (k=16)



Mean Shift (W=21)

Mean Shift Segmentation Results

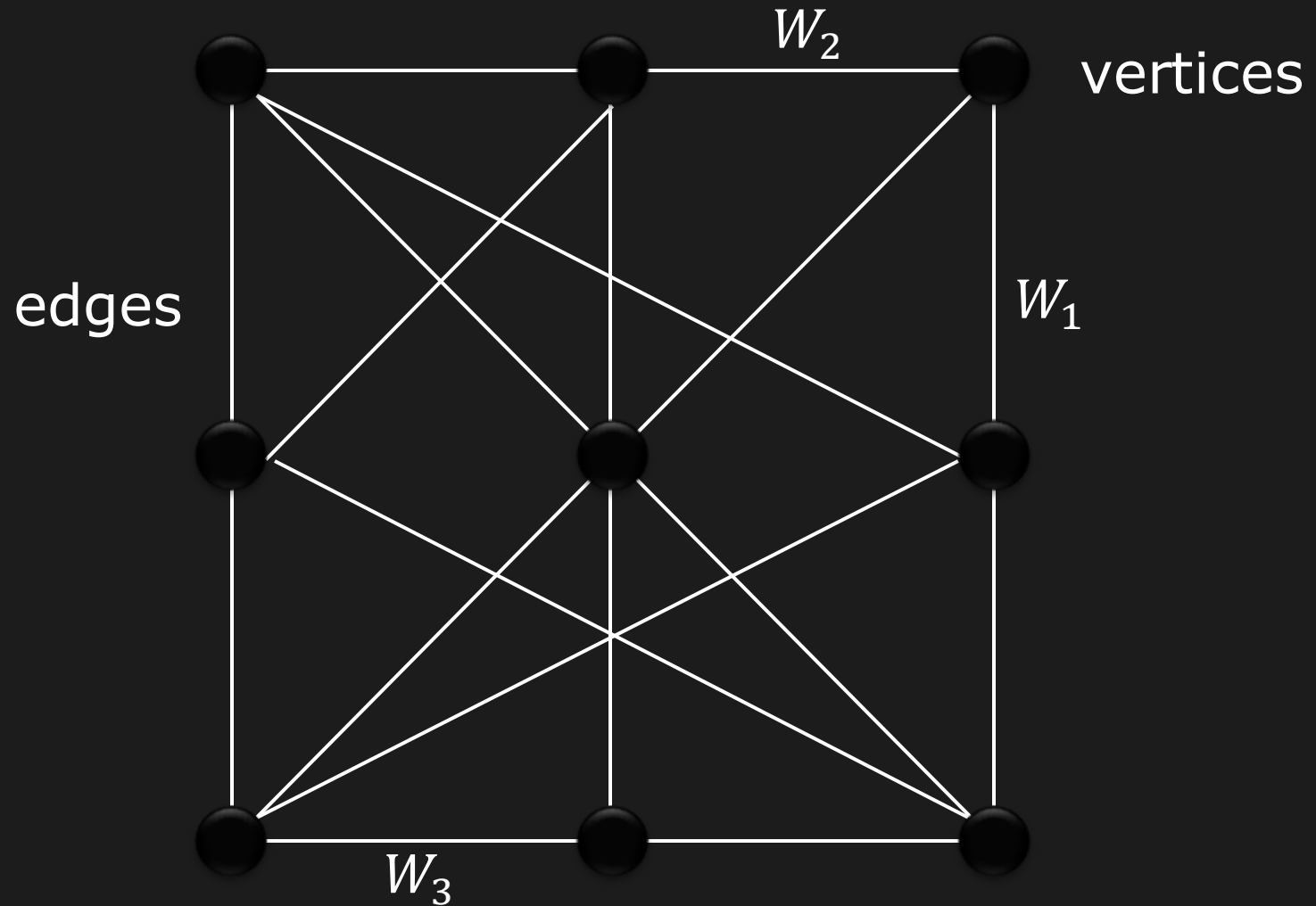


Examples of Mean Shift Segmentation

Mean Shift: Comments

- Simple but computationally expensive
- Finds arbitrary number of clusters
- No initialization required
- Clustering depends on window size W

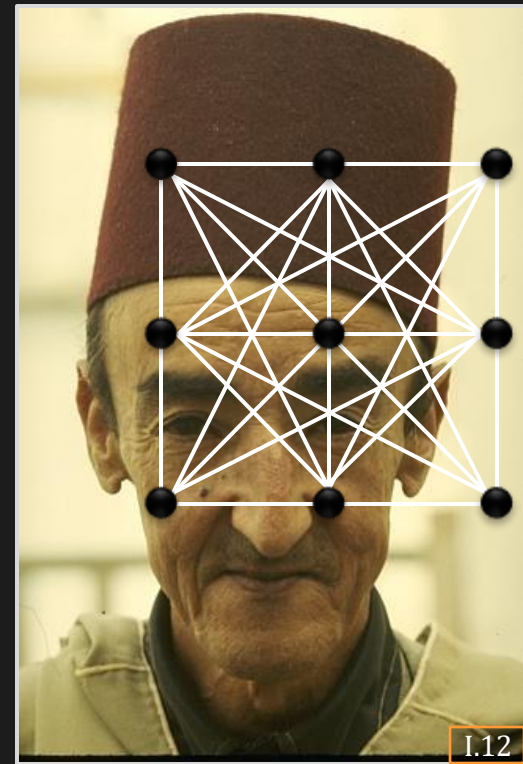
Graph Based Segmentation



Graph Based Segmentation

Images as Graphs:

- A vertex for each pixel.
- An edge between each pair of pixels.
- Graph Notation: $G = (V, E)$ where V and E are the sets of vertices and edges, respectively.
- Each edge is weighted by the **affinity** or similarity between its two vertices.



Input Image

Measuring Affinity

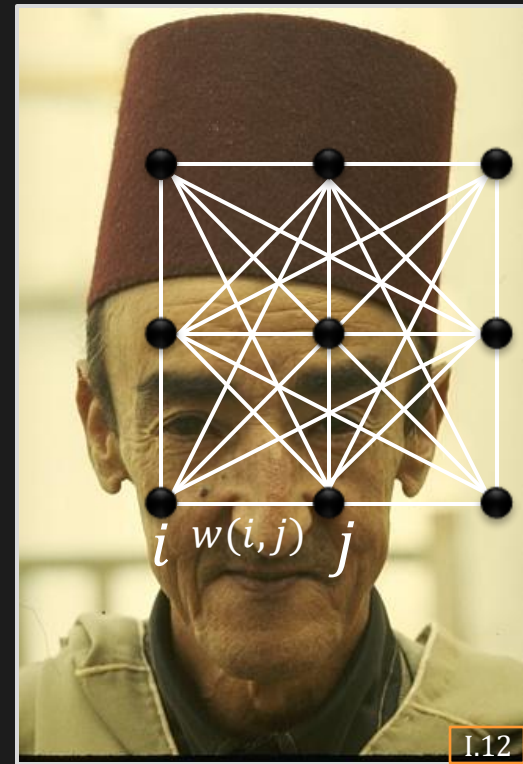
Let i and j be two pixels whose features are $\mathbf{f}_i$ and $\mathbf{f}_j$.

Pixel Dissimilarity:

$$S(\mathbf{f}_i, \mathbf{f}_j) = \sqrt{\left(\sum_k (f_{ik} - f_{jk})^2 \right)}$$

Pixel Affinity:

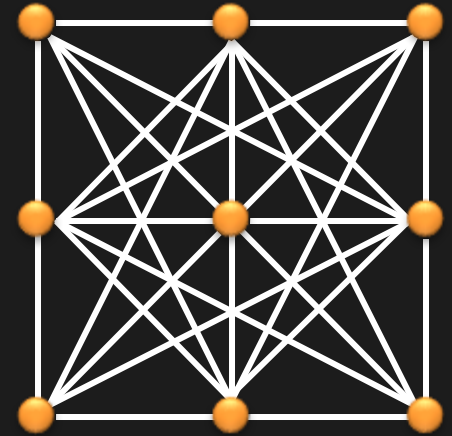
$$w(i, j) = A(\mathbf{f}_i, \mathbf{f}_j) = e^{\left\{ \frac{-1}{2\sigma^2} S(\mathbf{f}_i, \mathbf{f}_j) \right\}}$$



Input Image

Smaller the Dissimilarity, Larger the Affinity

Graph Cut



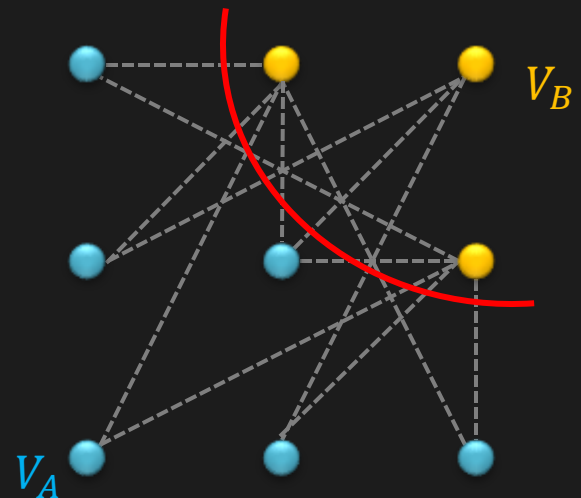
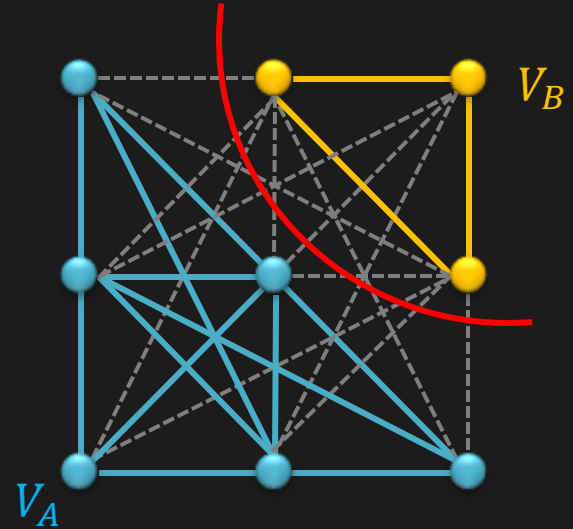
Graph Cut

Cut $C = (V_A, V_B)$ is a partition of vertices V of a graph $G = (V, E)$ into two **disjoint subsets** V_A and V_B .

Cut-Set: Set of edges whose vertices are in different subsets of partition.

Cost of Cut: Sum of weights of cut-set edges.

$$\text{cut}(V_A, V_B) = \sum_{u \in V_A, v \in V_B} w(u, v)$$

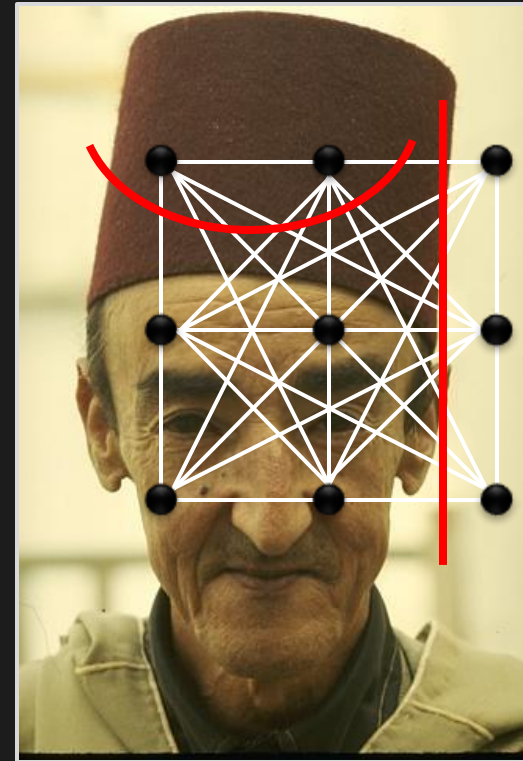


Graph Cut Segmentation

Criteria for Graph Cut:

- A pair of vertices (pixels) within a subgraph have **high affinity**.
- A pair of vertices from two different subgraphs have **low affinity**.

That is, minimize the cost of cut.
Also called **Min-Cut**.



Input Image

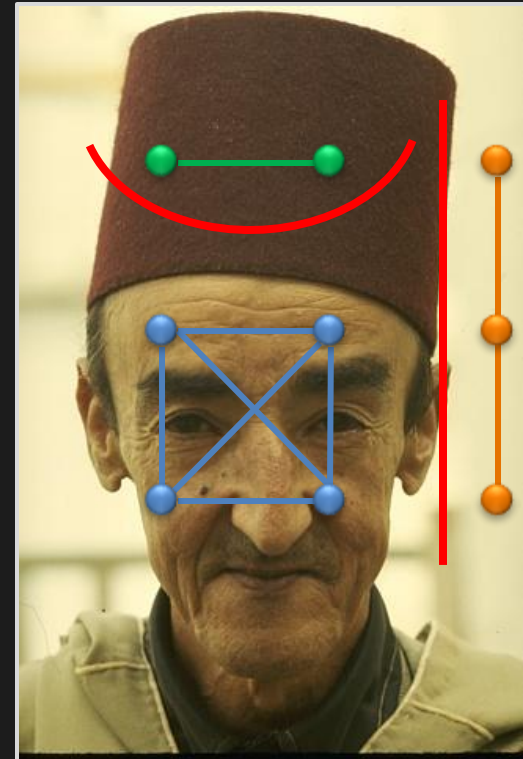
Graph Cut Segmentation

Criteria for Graph Cut:

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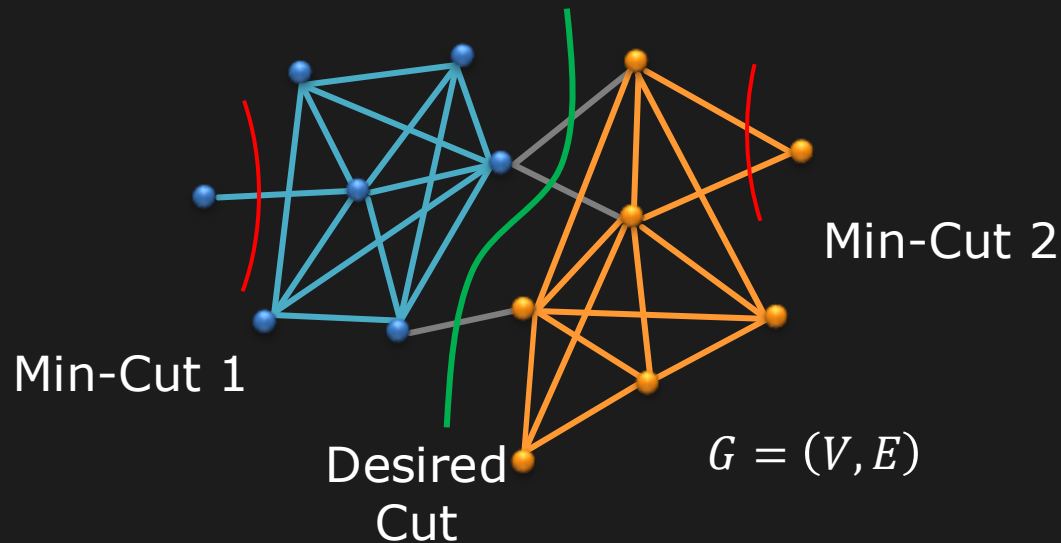
Each subgraph is an image segment.



Input Image

Problem with Min-Cut

There is a bias to cut small, isolated segments.

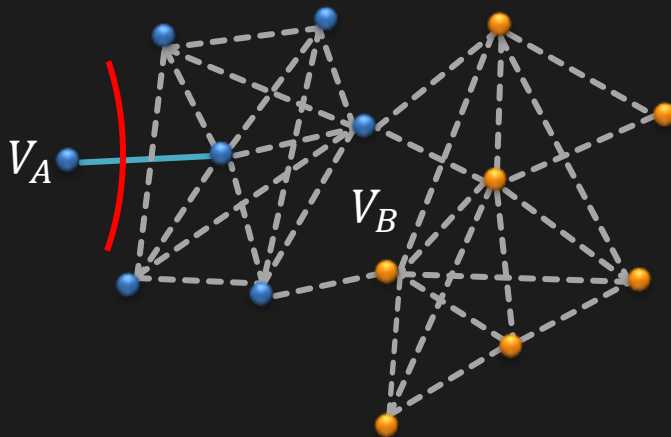


Solution: **Normalize Cut** to favor larger subgraphs.

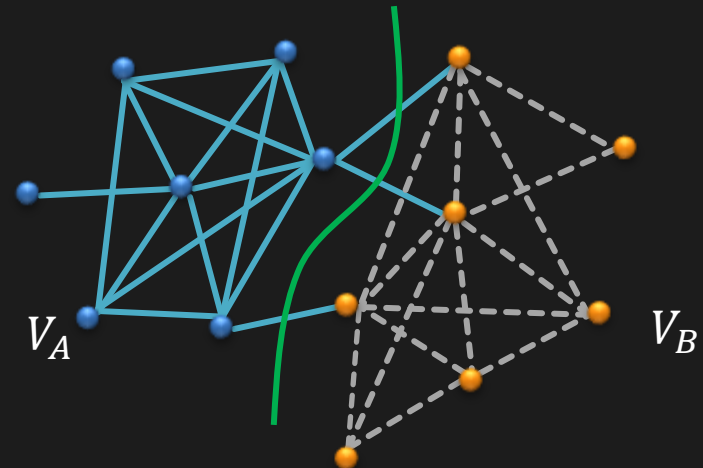
Measure of Subgraph Size

Compute how strongly vertices V_A are associated with vertices V .

$$assoc(V_A, V) = \sum_{u \in V_A, v \in V} w(u, v)$$



Weak $assoc(V_A, V)$



Strong $assoc(V_A, V)$

$assoc()$ is the sum of the weights of the solid edges

Normalized Cut (NCut)

Minimize Cost of Normalized Cut during Partition

$$NCut(V_A, V_B) = \frac{cut(V_A, V_B)}{assoc(V_A, V)} + \frac{cut(V_A, V_B)}{assoc(V_B, V)}$$

Minimizing NCut has no known polynomial time solution.
It is **NP-Complete**.

Fast eigenvector-based approximations exist.
See [Shi 2000].

NCut Segmentation Results



I.13

Segmented Images

Pixel Feature: $\{Brightness, Location\}$

Image Segmentation Summary

- Important for object detection and recognition.
- k-Means Segmentation
- Mean Shift Segmentation
- Normalized Cut Segmentation
- Many variations of mean-shift and graph based algorithms exist. [Paris 2007] [Wang 2004] [Tolliver 2006] [Cour 2005] [Felzenszwalb 2004] [Boykov 2004]

References: Textbooks

Computer Vision: Algorithms and Applications

Szeliski, R., Springer

Computer Vision: A Modern Approach

Forsyth, D and Ponce, J., Prentice Hall

References: Papers

[Comaniciu 2002] D. Comaniciu and P. Meer. "Mean Shift: A Robust Approach toward Feature Space Analysis". PAMI 2002.

[Fukunaga 1975] K. Fukunaga and L. D. Hostetler. "The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition" . IEEE Transactions on Information Theory 21. 1975.

[Kass 1987] M. Kass, A. Witkin and D. TerzoPoulos. "*Snakes: Active Contour Models*", *IJCV*, 1987.

[Lloyd 1957] S. P. Lloyd. "Least square quantization in PCM". IEEE Transactions on Information Theory 28. 1957.

[MacQueen 1967] J. B. MacQueen. "Some Methods for classification and Analysis of Multivariate Observations" . Proc of 5th Berkeley Symposium on Mathematical Statistics and Probability. University of California Press. 1967.

[Shi 2000] J. Shi and J. Malik. "Normalized cuts and image segmentation". PAMI 2000.

[Smith 1988] B. Smith. "Foundations of Gestalt Theory". Philosophia Verlag, 1988.

References: Papers

- [Boykov 2004] Y. Boykov and G. Funka-Lea. "Graph cuts and efficient N-D image segmentation." IJCV 2004.
- [Cour 2005] T. Cour, F. Benezit, and J. Shi. "Spectral segmentation with multiscale graph decomposition." CVPR 2005.
- [Felzenszwalb 2004] P. F. Felzenszwalb and D. P. Huttenlocher. "Efficient graph-based image segmentation." IJCV 2004.
- [Martin 2001] D. Martin, C. Fowlkes, D. Tal and J. Malik. "A Database of Human Segmented Natural Images and its Application to Evaluating Segmentation Algorithms and Measuring Ecological Statistics." ICCV 2001.
- [Paris 2007] S. Paris and F. Durand. "A topological approach to hierarchical segmentation using mean shift." CVPR 2007.
- [Todorovic 2008] D. Todorovic. "Gestalt principles." Scholarpedia, 3(12):5345.
- [Tolliver 2006] D. Tolliver and G. Miller. "Graph partitioning by spectral rounding: Applications in image segmentation and clustering." CVPR 2006.
- [Wang 2004] J. Wang, B. Thiesson, Y. Xu and M. Cohen. "Image and video segmentation by anisotropic kernel mean shift." ECCV 2004.

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- I.1 Marr, D. (1982). Vision. New York, NY: W. H. Freeman. p 101, Fig 3-1.
- I.2 <http://www.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>
- I.3 <http://www.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>
- I.4 Matlab demo image
- I.5 http://en.wikipedia.org/wiki/K-means_clustering#Standard_algorithm
- I.6 Matlab demo image
- I.7 D. Comaniciu and P. Meer. "Mean Shift: A Robust Approach toward Feature Space Analysis". PAMI 2002. Fig 2a.
- I.8 D. Comaniciu and P. Meer. "Mean Shift: A Robust Approach toward Feature Space Analysis". PAMI 2002. Fig 2c.
- I.9 D. Comaniciu and P. Meer. "Mean Shift: A Robust Approach toward Feature Space Analysis". PAMI 2002. Fig 2b.
- I.10 http://en.wikipedia.org/wiki/File:ClusterAnalysis_Mouse.svg
- I.11 D. Comaniciu and P. Meer. "Mean Shift: A Robust Approach toward Feature Space Analysis". PAMI 2002. Fig 10.
- I.12 <http://www.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>